LYLE OKOTH

SOFTWARE ENGINEER | FULL-STACK & VOICE-FIRST PRODUCTS

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years of experience shipping end-to-end web products — from problem framing and UI/UX design through backend APIs at 1.2M requests/day to deployment and feedback loops. Specialised in voice-first services: IVR and WhatsApp assistants serving 40,000+ monthly users, real-time speech streaming front-ends, and AI-assisted onboarding. Introduced Docker-based CI/CD that cut release lead time from 3 weeks to 2 days, and grew front-end test coverage from 0 to 71%, reducing reported production bugs 34%.

CORE COMPETENCIES

Backend: REST and WebSocket APIs, synchronous and asynchronous services for real-time streaming and batch processing, authentication & authorization, payment webhooks, USSD menus, Twilio voice/SMS

Frontend & Product: React interfaces for speech products (real-time translation, transcription, synthesis), UI/UX design, AI-assisted onboarding workflows, documentation that sped up new-hire onboarding

Platform & Reliability: Docker, GitHub Actions CI/CD, Kubernetes, logging and tracing for API-usage monitoring and downtime prevention, bot-abuse mitigation

ML Integration: Serving models behind FastAPI, LLM-powered features (onboarding, pre-labeling), RAG and agent workflows with LangChain / LlamaIndex

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C++, SQL

Frameworks: Django, FastAPI, Flask, Node.js, React

Data & Messaging: PostgreSQL, Redis, Elasticsearch, Apache Kafka, Celery

DevOps & Cloud: Docker, Kubernetes, GitHub Actions CI/CD, AWS, GCP, Render

Other: Twilio, payment integrations, WebSocket / WebRTC streaming, scikit-learn, Hugging Face

PROFESSIONAL EXPERIENCE

Software Engineer — SautiFlow Labs, Nairobi 2020 – 2022

- Shipped 5 production features for an IVR and WhatsApp voice assistant serving 40,000+ monthly users, from requirement framing with non-technical clients through design, build and iteration.
- Built REST and WebSocket APIs in Django and Node.js handling 1.2M requests/day while holding p95 latency under 140 ms.
- Introduced Docker deployments and a GitHub Actions CI/CD pipeline: release cadence moved from monthly to weekly and lead time dropped from 3 weeks to 2 days.
- Took front-end test coverage from 0 to 71%; reported production bugs fell 34% over two quarters.
- Trained and served the team's first machine-learning model — a scikit-learn intent classifier behind FastAPI that deflected 28% of support chats; integrated Twilio voice, SMS, payment webhooks and USSD menus.
- Reviewed junior pull requests and ran the internal weekly code club.

Machine Learning Engineer (full-stack responsibilities) — SautiFlow Labs, Nairobi 2024 – Present

- Worked with software engineers to design and ship web interfaces for speech products including real-time streaming translation, transcription and synthesis — contributing to high retention rates.

- Deployed model services (FastAPI, Triton, Kubernetes) serving streaming speech at 310 ms median first-audio latency; instrumented logging and tracing to monitor token usage and detect model drift.
- Built robust authentication and authorization controls that prevented API misuse by bots, improving the experience for real users.

Data Engineer — SautiFlow Labs, Nairobi 2022 – 2024

- Built a fast, scalable synchronous and asynchronous backend for real-time and batch processing of user requests, plus the data studio for human + AI collaborative data collection and annotation.
- Designed pipelines (Airflow, Kafka, Spark) processing 6 TB of audio per year; wrote detailed documentation that made onboarding new hires easier.

SELECTED PROJECTS

SautiFlow — Script-to-Skit Studio — Turns a written script into a finished skit: casting, voicing, timing and mix. Creators type the joke, pick a cast, and get back ten renderable variations instead of one take they have to record themselves — full product surface from web UI to render backend.

LabelSet — Speech Data Platform — Web application for sourcing recordings, collaborative human + AI labelling, and label-quality verification — the pipeline behind every model on the portfolio.

FaceForm — Persona Generator — Generates consistent, clearly synthetic faces for teams that need people in their visuals without licensing real ones; a designer sets an archetype once, then draws a whole persona family from the same latent noise.

PigaMagic — Event Photo Gallery — Takes a few phone snapshots per guest at a 200-person event and returns a gallery of polished event photos for every guest, not just the ones near the camera.

EDUCATION

BSc, Computer Security and Forensics — Jaramogi Oginga Odinga University of Science and Technology (JOOUST) 2016 – 2020

Diploma, Mechatronics Engineering — JKUAT, Juja, Kenya 2013 – 2016

CERTIFICATIONS

AWS, Google Cloud, DeepLearning.AI and NVIDIA DLI certifications (2021 – 2025).

LEADERSHIP

- Co-Founder — SautiFlow Labs (2022)
- Volunteer — OMDENA (2024)
- Founded a 12-member coding club and ran weekend build sessions

REFERENCES

Available on request.